



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
PRE-SCORE
08-10-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

Time : 3 Hours

PAPER-2 (OPTIONAL)

Maximum Marks : 180

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

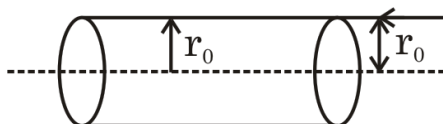
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. I is the image of a point object O formed by spherical mirror then which of the following statement(s) is/are **CORRECT**?
- (A) If O and I are on same side of the principal axis, then they have to be on opposite side of the mirror
- (B) If O and I are on opposite sides of the principal axis, then they have to be on same side of the mirror
- (C) If O and I are on opposite side of the principal axis, then they can be on opposite side of the mirror as well.
- (D) If O and I are on same side of the principal axis, then they have to be on same side of the mirror

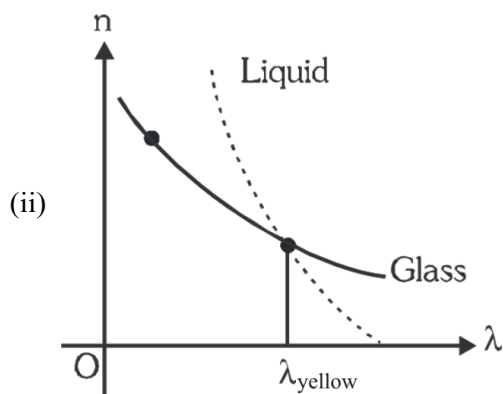
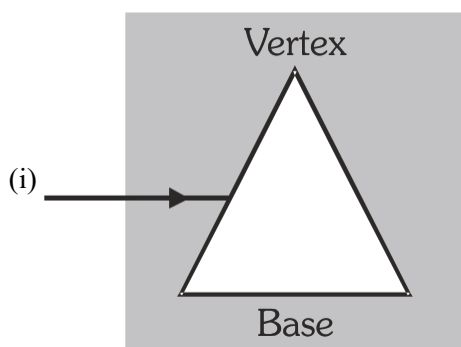
2. A solid transparent cylinder is made of a material whose refractive index depends on the distance r from its axis of symmetry according to the formula

$$n = 1 + (n_0 - 1) \cdot \left(1 - \left(\frac{r}{r_0} \right)^2 \right)$$

Here r_0 is the radius of the cylinder, $n_0 = 5/4$. A narrow laser beam falls perpendicular to the end of the cylinder in the direction parallel to the axis of the cylinder on a point located at a distance of r_0 from the axis of cylinder. Choose the correct statement(s) relating to the path followed by the narrow beam within the cylinder

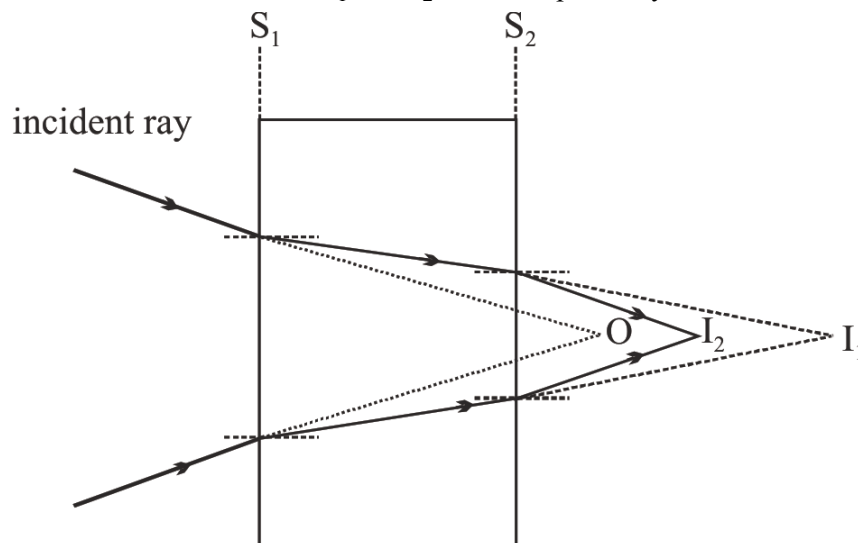


- (A) The beam follows a helical path around the axis of the cylinder
- (B) The beam follows a path in a plane containing the axis of the cylinder and its initial direction of propagation
- (C) The path crosses the axis at a distance of $5r_0 \ln 3$ from the end of which beam is incident
- (D) The path crosses the axis at a distance of $4r_0 \ln 2$ from the end on which the beam is incident
3. A glass prism is immersed in a hypothetical liquid as shown in figure (i). The curves showing the refractive index n as a function of wavelength λ for glass and liquid. When a ray of white light is incident on the prism parallel to the base



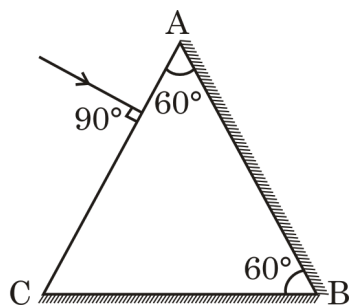
- (A) yellow ray travels without deviation
- (B) blue ray is deviated towards the vertex
- (C) red ray is deviated towards the base
- (D) there is no dispersion

4. Figure shows converging rays on a glass slab placed in air. O is point of intersection of incident rays. I_1 and I_2 are image formed after refraction at S_1 and S_2 surfac respectively. Mark the correct option(s)

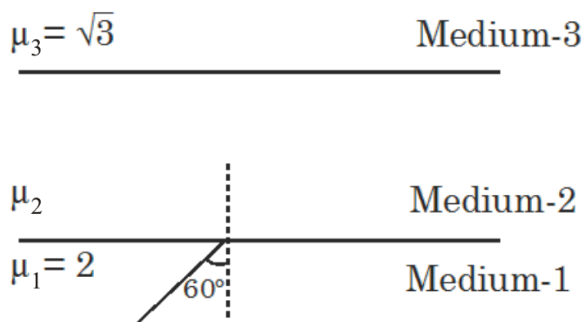


- (A) for refraction at S_2 , I_1 is real object and I_2 is real image
- (B) for refraction at S_1 , O is real object and I_1 is its real image.
- (C) for refraction at surface S_1 , O is virtual object and I_1 is real image
- (D) for refraction at surface S_2 , I_1 is virtual object and I_2 is real image

5. The two faces of a prism of side length a are silvered. A ray incident normally on the unsilvered face as shown. Then



- (A) net deviation of ray through prism is 90°
 (B) net deviation of ray through prism is 180°
 (C) length of path travelled by light ray inside the prism is $\sqrt{5}a$
 (D) length of path travelled by light ray inside the prism is $\sqrt{3}a$
6. A light ray is incident on a boundary as shown.



- (A) If $\mu_2 < \sqrt{3}$, light will enter medium-3 with deviation 30° .
 (B) If $\mu_2 > \sqrt{3}$, light will enter medium-3 with deviation 30° .
 (C) If $\mu_2 > \sqrt{3}$, light will never return to medium-1 again.
 (D) If $\mu_2 < \sqrt{3}$, light will again return to medium-1.

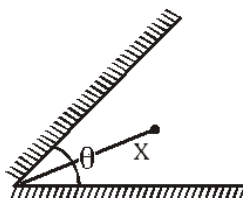
SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

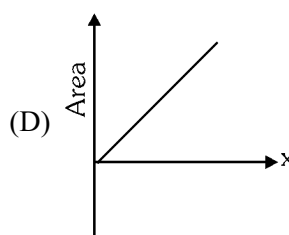
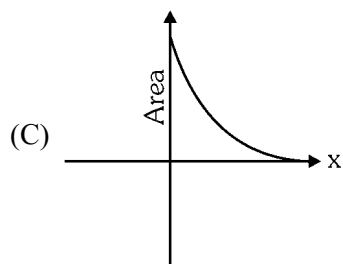
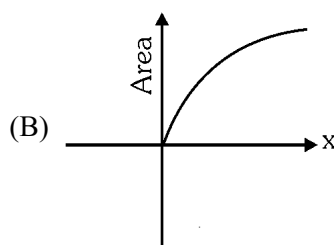
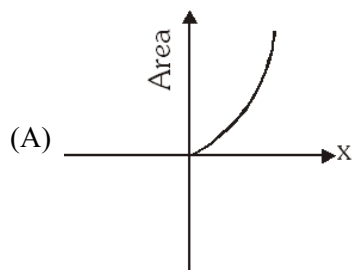
Full Marks : +3 If ONLY the correct option is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
 Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Whenever an object is placed between the two mirrors at an angle θ on its bisector. It forms several images which lies on a circle of radius equal to the distance between the vertex of mirror and the object.



7. Draw graph between the area formed by joining all the images and object and the distance between the vertex and object



8. If the angle between the two mirror is 60° . Find the area of regular polygon bounded by the images and the object

(A) $\frac{x^2 (4\sqrt{3} + 1)}{4}$

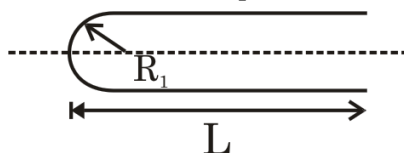
(B) $x^2 \sqrt{\left(\frac{4\sqrt{3} + 1}{4}\right)} x$

(C) $\frac{3\sqrt{3}x^2}{2}$

(D) $\frac{5\sqrt{3}x^2}{4}$

Paragraph for Questions 9 and 10

There is a glass rod placed in the air whose refractive index $n = 1.5$, the central axis has length $L = 45\text{cm}$, one end is a convex spherical surface with a radius of $R_1 = 10\text{cm}$. This surface faces object. We want to make the function of the glass rod equivalent to that of an ideal astronomical telescope (magnifying the object at infinity on the main optical axis so that the final image is formed at infinity on the main optical axis). Take the central axis of the glass rod as the main optical axis.



9. What kind of spherical surface should the other end be?
- (A) Concave with radius 5 cm.
(B) Convex with radius 5 cm.
(C) Concave with radius 10 cm.
(D) Convex with radius 20 cm.
10. What is the angular magnification of this system for the object at infinity?
- (A) 4
(B) 2
(C) 3
(D) 6

SECTION-II (i) : (Maximum Marks: 12)

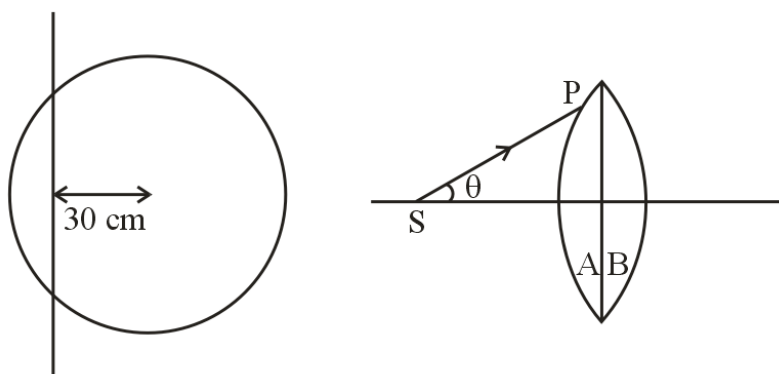
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

There are two spheres of transparent materials. Sphere A has refractive index of $\mu = \frac{8}{5}$ and refractive index of sphere B is not known to us (consider it as μ_2). Radii of each sphere is 50 cm. Both the spheres are cut so that the section of cut is 30 cm from center. Later the smaller parts struck together as shown.

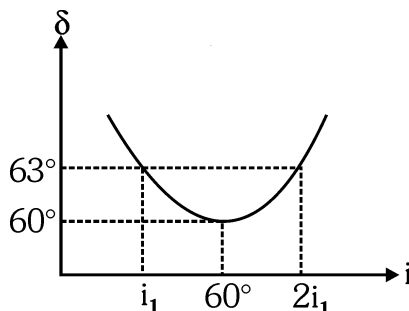


A narrow laser beam from point S (source) is incident on the curved surface of the part of sphere A at point P. The length of beam from S to P is 50 cm. Also the angle made by beam with the principal axis is $\theta = 37^\circ$ ($\sin \theta = \frac{3}{5}$). The beam finally emerges from the curved surface of the part of sphere B. The beam follows a total deviation of 53° .

1. If the image of first refraction is formed at a distance D cm from the surface of A, find the value of $\frac{1}{D}$.
2. If the beam after emerging from B meets the principal axis at I, the total length of beam (in cm) from S to I will be given by $\frac{10}{7}Q$. Find the value of Q.

Paragraph for Questions 3 and 4

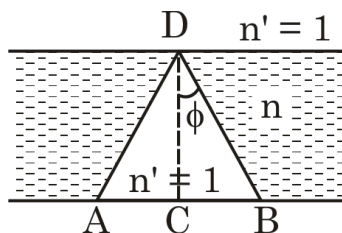
For a glass prism the plot of deviation (δ) vs angle of incidence (i) is as shown. If the refractive index of the glass is μ and value of angle is i_1 . Then :



3. Find the value of i_1 (in degree) :
4. Find the refractive index of glass :

Paragraph for Questions 5 and 6

As shown in the figure, there is large vessel filled with a liquid with a refractive index n , and a symmetrical cone-shaped thin-walled transparent cover ADB is stuck in the middle of the vessel. The top angle is 2ϕ , inside of the cone is filled with air, the whole cover is submerged in the liquid, and there is a bright spot C at the midpoint of AB at the bottom of the tank. Take $n = 5/3$.



5. What is the minimum value of $\tan \phi$ so that an observer outside may be able to see the light coming from C?
6. If $\phi = 87^\circ$, what will be the angle (in degree) between the extreme rays perceived by an observer just above D?

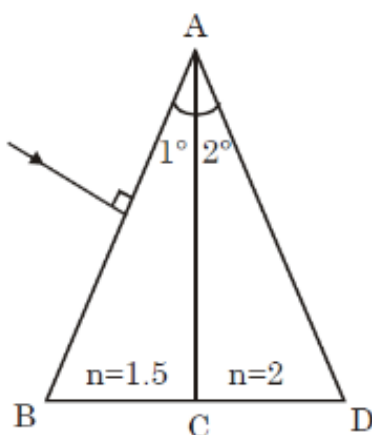
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. A ray of light is incident on a prism as shown in the figure. The total deviation suffered by the ray is $\frac{x^0}{2}$. Then find x. ($\angle BAC = 1^\circ$ and $\angle CAD = 2^\circ$, refractive indices are shown in figure.)



8. On the surface of a liquid reservoir of depth 3m is a circular raft of radius 6m. The surface is illuminated by light incident from all the directions. Determine the radius (in m) of the shadow of the raft at the bottom of the reservoir. The refractive index of the liquid is 1.25.
9. A point object is placed at a distance of 25 cm from a convex lens of focal length 20 cm. If a glass slab of thickness t and refractive index 1.5 is inserted between the lens and object. The image is formed at infinity. Find the thickness t (in cm)?

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

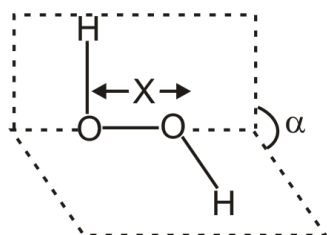
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

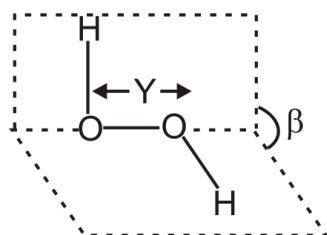
Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which of the following option(s) is/are correct for the structure of H_2O_2 in gas phase and solid phase?



(a) Gas phase



(a) Solid phase

- (A) $X > Y$ (O – O) bond length
(B) $X < Y$ (O – O) bond length
(C) $\alpha > \beta$ (dihedral angle)
(D) $\alpha < \beta$ (dihedral angle)

2. Which of the following property is not same of both H_2 and D_2 ?
- (A) Enthalpy of fusion
(B) Enthalpy of vaporization
(C) Internuclear distance
(D) Enthalpy of bond dissociation
3. The used organic solvent in industrial preparation of H_2O_2 must
- (A) Dissolve quinol
(B) Dissolve quinone
(C) Resist oxidation
(D) Be immiscible with water
4. Lanthanoids (Ln) when,
- (A) Heated with S, produces Ln_2S_3
(B) Burns in O_2 , produces Ln_2O_3
(C) React with H_2O produces $Ln(OH)_3 + H_2$
(D) React with halogens, produces LnX_3
5. Select the correct matching of maximum prescribed concentration in drinking water with corresponding metals.
- (A) Fe – 0.2 mg dm^{-3}
(B) Cu – 0.05 mg dm^{-3}
(C) Zn – 5 mg dm^{-3}
(D) Cd – 0.2 mg dm^{-3}
6. The pair(s) of complexes where in both exhibit tetrahedral geometry is(are)
- (A) $[FeCl_4]^-$ and $[Fe(CO)_4]^{2-}$
(B) $[Co(CO)_4]^-$ and $[CoCl_4]^{2-}$
(C) $[Ni(CO)_4]$ and $[Ni(CN)_4]^{2-}$
(D) $[Cu(py)_4]^+$ and $[Cu(CN)_4]^{3-}$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
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Full Marks : +3 If **ONLY** the correct option is chosen.

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Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Structural isomerism arises due to the difference in the type of chemical linkages and distribution of ligands within and outside the coordination sphere whereas stereoisomers differ only with respect to the spatial orientation of ligands in space around the metal ion.

7. Select the type of complex which have the ratio of total number of enantiomeric pairs to the total number of stereoisomers equal to $\frac{3}{4}$.
- (A) $[M(AB)_2bc]$
- (B) $[M(AB)_2a_2]$
- (C) $[Ma_2b_2cd]$
- (D) $[M(AB)_2ab]$
8. In the complex $[Pt(Cl)(NO_2)(NO_3)(SCN)]^{2-}$,
- Total number of possible isomers = a
- Total number of unpaired electrons on central metal ion = b
- Number of inner/outer d-orbitals involved in the hybridisation of the complex = c
- The value of $(a \times c) + b$ is :-
- (A) 12
- (B) 13
- (C) 3
- (D) 0

Paragraph for Questions 9 and 10

Co-ordination compounds are used as catalyst for many industrial processes. Example include Rhodium complex used for the hydrogenation of alkenes, this catalyst is called as Wilkinson catalyst.

9. What is true about Wilkinson catalyst?
- (A) Co-ordination number of Rhodium is three
 (B) Oxidation number of Rhodium is +2
 (C) Co-ordination number of Rhodium is six
 (D) Oxidation number of Rhodium is +1
10. There are four different compounds of Co^{3+} with ligand Cl^- , NH_3 , H_2O and CN^- that are A, B, C and D. Consider the following information.

Compound	Wavelength of light absorbed (nm)
A	535
B	500
C	475
D	310

The possible combination of A, B, C and D

- (A) $\text{A} \rightarrow [\text{CoCl}(\text{NH}_3)_5]^{2+}$, $\text{B} \rightarrow [\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]^{3+}$, $\text{C} \rightarrow [\text{Co}(\text{NH}_3)_6]^{3+}$, $\text{D} \rightarrow [\text{Co}(\text{CN})_6]^{3-}$
 (B) $\text{A} \rightarrow [\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]^{3+}$, $\text{B} \rightarrow [\text{CoCl}(\text{NH}_3)_5]^{2+}$, $\text{C} \rightarrow [\text{Co}(\text{NH}_3)_6]^{3+}$, $\text{D} \rightarrow [\text{Co}(\text{CN})_6]^{3-}$
 (C) $\text{A} \rightarrow [\text{Co}(\text{CN})_6]^{3-}$, $\text{B} \rightarrow [\text{Co}(\text{NH}_3)_6]^{3+}$, $\text{C} \rightarrow [\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]^{3+}$, $\text{D} \rightarrow [\text{CoCl}(\text{NH}_3)_5]^{2+}$
 (D) $\text{A} \rightarrow [\text{CoCl}(\text{NH}_3)_5]^{2+}$, $\text{B} \rightarrow [\text{Co}(\text{CN})_6]^{3-}$, $\text{C} \rightarrow [\text{Co}(\text{NH}_3)_6]^{3+}$, $\text{D} \rightarrow [\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]^{3+}$

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

All the lanthanoids are silvery white soft metals and tarnish rapidly in air. They have typical metallic structure and are good conductors of heat and electricity.

1. The atomic number of element having unusually large melting point than the other lanthanides is :-
2. Consider the electronic configuration of lanthanides series. If the unpaired electrons in the ground state of Ce^{+3} , Yb^{+3} and Gd^{+3} are a, b and c respectively, then the value of $a+b+c$?

Paragraph for Questions 3 and 4

Hydrogen peroxide is an important chemical used in pollution control treatment of domestic and industrial effluents.

3. Consider the following species :-
 Fe^{2+} , PbS , MnO_4^- , HOCl , Mn^{2+} , I_2
 Number of species which shows reducing action with H_2O_2 in acidic medium with their valence factor '2' = a
 Number of species which shows oxidizing action with H_2O_2 in acidic medium with their valence factor '8' = b
 Number of species which shows reducing action with H_2O_2 in basic medium with their valence factor '3' = c
 Calculate $\frac{b}{a+c}$.
4. If commercially marketed sample of H_2O_2 solution is 4.54 V, then % H_2O_2 in the solution is :-

Paragraph for Questions 5 and 6

In 1980's atmospheric scientists working in Antarctica reported about depletion of ozone layer commonly known as ozone hole over the south pole. It was found that a unique set of conditions was responsible for the ozone hole.

5. How many of the following species (free radicals and molecules) are responsible for the depletion of ozone layer?
 CF_2Cl_2 , $\text{C}\ell\text{ONO}_2$, CH_4 , $\text{C}\ell\text{O}$, CH_3 , Cl , CH_2O , HOCl , NO_2 , HCl
6. When chlorine free radical reacts with methane in summer season, the free radical product formed have total 'x' number of hybrid orbitals. Find 'x'.

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

-
7. A hydrated isomer of $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$, for which no geometrical isomer is possible, when 0.1 mole of such complex in 1 L solution is treated with excess of AgNO_3 , mass of white ppt. formed is :-
(Give nearest rounded off integer)
8. In Mischmetall, ratio of (approx.) % Lanthanoid metal used to (approx.) % iron used is :-
9. Find out the number of following orders which are **INCORRECT** against the mentioned properties :
- (i) Ortho hydrogen > Para hydrogen (Stability at low temperature)
 - (ii) Ethanol > Glycerol (Viscosity)
 - (iii) $\text{D}_2 < \text{He}$ (Boiling Point)
 - (iv) $\text{HF} > \text{H}_2\text{O}$ (Melting point)
 - (v) $\text{H}_3\text{BO}_3 > \text{BF}_3$ (Melting point)
 - (vi) $\text{NH}_3 < \text{SbH}_3$ (Boiling point)
 - (vii) $\text{H}_2\text{O}_2 > \text{H}_2\text{O}$ (Strength of hydrogen bond)
 - (viii) $\text{H}_2\text{O} < \text{D}_2\text{O}$ (Freezing point)
 - (ix) $\text{HF}_2^\ominus < \text{HF}$ (Strength of hydrogen bond)
 - (x) $\text{D}_2\text{O} < \text{H}_2\text{O}$ (Bond energy)

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Letters of the word "UNIVERSITY" are arranged randomly. Then

- (A) Probability that both 'I' do not come together is $4/5$
- (B) Probability that all vowels are separated is $1/6$
- (C) Probability that all vowels are together is $1/30$
- (D) Probability that word starts and ends with a consonant is $1/3$

2. Let $\vec{p} = \vec{a} \times (\vec{b} + \vec{c})$, $\vec{q} = \vec{b} \times (\vec{c} + \vec{a})$ and $\vec{r} = \vec{c} \times (\vec{a} + \vec{b})$. Then

- (A) $\vec{p}, \vec{q}, \vec{r}$ are coplanar
- (B) If $x\vec{p} + y\vec{q} + z\vec{r} = \vec{0}$; $x, y, z \in \mathbb{R}$ then $\frac{x+z}{y} = 2$
- (C) $\vec{p} \cdot \vec{q} + \vec{q} \cdot \vec{r} + \vec{r} \cdot \vec{p} = -(|\vec{p}|^2 + |\vec{q}|^2 + |\vec{r}|^2)$
- (D) $\vec{p} \cdot \vec{r} = \frac{1}{2}(|\vec{q}|^2 - |\vec{p}|^2 - |\vec{r}|^2)$

3. Sides a, b, c of $\triangle ABC$ are roots of the equation $3x^3 - px^2 + qx - 4 = 0$ ($p, q \in \mathbb{R}$) such that $\frac{a}{3} + \frac{b}{4} + \frac{c}{3} = 1$. Then
- (A) $\triangle ABC$ is equilateral
- (B) $\triangle ABC$ is isosceles
- (C) Area of $\triangle ABC = \frac{2\sqrt{5}}{9}$
- (D) Circum radius of $\triangle ABC = \frac{3}{10}$
4. A ray is sent along the line $\frac{x}{2} = \frac{y-2}{2} = \frac{z-1}{0}$ and is reflected by the plane $x = 0$ at point P. The reflected ray is again reflected by the plane $x + 2y = 0$ at point Q. The initial ray and final reflected ray meet at point R. Then
- (A) The co-ordinates of Q is $(4, -2, 1)$
- (B) The co-ordinates of R is $(-3, -1, 1)$
- (C) The centroid of $\triangle PQR$ is $(0, 0, 0)$
- (D) The centroid of $\triangle PQR$ is $\left(\frac{1}{3}, \frac{-1}{3}, 1\right)$
5. A bag contains some white and some black balls, all combinations being equally likely. The total number of balls in the bag is 12. Four balls are drawn from the bag without replacement.
- (A) Probability that all balls are black is $1/5$
- (B) Probability that all balls are black is $1/3$
- (C) Probability that two balls are black and two are white is $1/4$
- (D) Probability that two balls are black and two are white is $1/5$
6. Which of the following statement is/are true ?
- (A) If $\vec{a}, \vec{b}, \vec{c}$ are co-terminus edges of a regular tetrahedron then $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a}$
- (B) If $\vec{a}, \vec{b}$ are two unit vector inclined at $\frac{\pi}{3}$, then $[\vec{a} \quad \vec{b} + \vec{a} \times \vec{b} \quad \vec{b}] = \frac{3}{4}$
- (C) Three points having position vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are collinear if $[\vec{a} \quad \vec{b} \quad \vec{c}] = 0$
- (D) If $\vec{a} \times \vec{b} = \vec{c}$ and $\vec{b} \times \vec{c} = \vec{a}$ then $\vec{a} \cdot \vec{b} = 0$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
 Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Let B_n denotes the event that n fair dice are rolled once with $P(B_n) = \frac{1}{2^n}$; $n \in \mathbb{N}$. Hence $B_1, B_2, B_3, \dots, B_n$ are pairwise mutually exclusive events as $n \rightarrow \infty$. The event E occurs with atleast one of the event $B_1, B_2, \dots, B_n$ and denotes that the sum of numbers appearing on dice is S .

7. Probability that greatest number on dice is 4, if three dice are known to have been rolled, is :

(A) $\frac{37}{216}$ (B) $\frac{64}{216}$ (C) $\frac{27}{216}$ (D) $\frac{31}{216}$

8. If $S = 3$, then $P\left(\frac{B_2}{S}\right)$ is equal to -

(A) $\frac{8}{169}$ (B) $\frac{24}{169}$ (C) $\frac{72}{169}$ (D) $\frac{16}{169}$

Paragraph for Questions 9 and 10

Lines $\frac{x-3}{2} = \frac{y-2}{3} = \frac{z-1}{\lambda}$ and $\frac{x-2}{3} = \frac{y-3}{2} = \frac{z-2}{3}$ intersect at point P. Then -

9. Equation of the plane containing both the lines is

(A) $x + 5y - 3z = 10$ (B) $x + 6y + 5z = 20$
 (C) $x + 6y - 5z = 10$ (D) $x + 5y + 3z = 20$

10. The distance of point P from the line $\frac{x-1}{2} = \frac{y-3}{4} = \frac{z-2}{3}$ measured parallel to the plane $3x + 2y - 2z + 15 = 0$ is -

(A) $\frac{17}{4}$ (B) $\frac{3\sqrt{21}}{4}$ (C) $\frac{3\sqrt{21}}{2}$ (D) $\frac{17}{2}$

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

On a normal standard die one of the 21 dots from any one of the six faces is removed at random with each dot equally likely to be chosen. The die is then rolled.

1. Probability of die showing even number of dots is $\frac{a}{b}$, where a & b are co-prime, then value of a + b is
2. Probability of die showing odd number of dots is $\frac{p}{q}$, where p and q are co-prime, then value of $q^2 - p^2$ is -

Paragraph for Questions 3 and 4

In a $\triangle ABC$, the incircle touches sides BC, CA and AB at D, E, F respectively such that AF, BD, CE are consecutive even integers and inradius is $\frac{4\sqrt{6}}{3}$.

3. If R is circum radius then find value of $\frac{24}{25}R^2$.
4. If ℓ_1, ℓ_2, ℓ_3 are length of medians then find value of $\ell_1^2 + \ell_2^2 + \ell_3^2$.

Paragraph for Questions 5 and 6

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-coplanar unit vector such that the angle between every pair of them is $\frac{\pi}{3}$. If $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} = p\vec{a} + q\vec{b} + r\vec{c}$, where p, q, r are scalars, then

5. The value of $[\vec{a} \ \vec{b} \ \vec{c}]^4$ is _____
6. The value of $20p^2 + 21q^2 - 22r^2$ is _____

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

-
7. Let position vectors of vertices of ΔABC be $3\hat{i} + 4\hat{j} + 5\hat{k}$, $-2\hat{i} + 6\hat{j} + \sqrt{10}\hat{k}$, $5\hat{j} - 5\hat{k}$. Let DEF be triangle formed by joining the mid points of sides of ΔABC . If $\vec{P}$ is the position vector of orthocentre of ΔDEF and position vector of Q is $2\hat{i} - 3\hat{j} + 6\hat{k}$, then find $|\vec{PQ}|$.
8. S is a set containing 5 elements. Two subsets A and B of set S are selected at random. The probability that $A \cap B$ contains 2 elements is P, then find value of $(512)P$
9. Let $\vec{a}$, $\vec{b}$, $\vec{c}$ are three non-zero, non-collinear vectors and $\vec{r}_1 = p\vec{a} + q\vec{b} + \vec{c}$ and $\vec{r}_2 = \vec{a} + p\vec{b} + q\vec{c}$. If the vectors $\vec{r}_1 + 2\vec{r}_2$ and $2\vec{r}_1 + \vec{r}_2$ are collinear then find value of $p^2 + q^2$.

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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Registered & Corporate Office : 'SANKALP', CP-6, Indra Vihar, Kota (Rajasthan) INDIA-324005

Ph. : +91-744-3556677, +91-744-2757575 | E-mail : info@allen.in | Website : www.allen.ac.in

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